| Question | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| 5 | (a) | | $\frac{\text{speed of light in a vacuum}}{\text{speed of light in the material}}$<br>Do not accept $n = \frac{c}{v}$ unless letters are clearly defined.<br>Accept answer based on ratio of sines if surrounding material stated as vacuum / $n$ = 1 / accept air for vacuum | 1 | | | 1 | | |
| | (b) | (i) | Substitution into: $n_1 \sin\theta_1 = n_2 \sin\theta_2$ i.e. $1.61 \sin 65 = n_2 \sin 80$ (1)<br>$n_2 = 1.48$ (1) | 1 | 1 | | 2 | 2 | |
| | | (ii) | Distance travelled = $\frac{10}{\cos 80}$ = 58 [mm] (1)<br>$\frac{3\times10^8}{1.48\ \mathbf{ecf}} = 2.03 \times 10^8$ [m s$^{-1}$] (1)<br>$\frac{58\times10^{-3}}{2.03\times10^8} = 2.86 \times 10^{-10}$ [s] (1) | | 3 | | 3 | 3 | |
| | (c) | (i) | $\sin \mathrm{C} = \frac{1.31}{1.48\ \mathbf{ecf}}$ (1)<br>C = 62.3° (1)<br>$< 80° \therefore$ TIR (1)<br>**Accept**<br>$n_1 \sin\theta_1 = n_2 \sin\theta_2$ with 80° (1)<br>Reports error message (1)<br>Hence no refraction therefore TIR (1) | 1<br>1 | 1 | | 3 | 2 | |
| | | (ii) | Light reflected at Y with angle $i$ = angle $r$ by eye | | 1 | | 1 | | |
| | | | **Question 5 total** | **4** | **6** | **0** | **10** | **7** | **0** |